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Department of Computer Science

CS11021– Programming Laboratory (Intake 43)

Lab Sheet 12

Part - I

Question 1 – Student Marks Management System

Write a C++ program for a **Student Marks Management System** using a menu-driven approach. The program should allow:

1. Enter marks for a student
2. Calculate and display total and average marks
3. Display grade (e.g., Pass/Fail based on average)
4. Exit

Requirements:

- Use variables only (no arrays).
- The program should repeat using a loop until the user exits.
- Handle invalid marks (e.g., negative values).

Question 2 – Simple Calculator System

Create a **menu-driven calculator program** in C++ that performs:

1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit

Requirements:

- Take two numbers as input for calculations.
- Do not allow division by zero.
- Use a loop to display the menu repeatedly until exit is selected.
- No arrays allowed.

Question 3 – Employee Salary System

Develop a C++ program for an **Employee Salary Management System** using a menu-driven approach. The program should include:

1. Enter basic salary
2. Calculate net salary (include simple allowance and deduction)
3. Display salary details
4. Exit

Requirements:

1. Use only variables (no arrays or multiple employees).
2. Use a loop to keep the program running until exit.
3. Show appropriate messages if salary is not yet entered.

Part II

1. Write a C++ program to:
    - Declare an integer array of size 5
    - Input 5 elements from the user
    - Display all elements

2. Write a C++ program to:
    - Store 6 numbers in an array
    - Print all elements using a loop
    - Display them in a single line

3. Write a C++ program to:
    - Input 5 integers into an array
    - Ask the user to enter a number to search
    - Display whether the number is found or not

4. Write a C++ program to:
    - Input 5 numbers into an array
    - Find and display the **maximum** and **minimum** values

5. Write a C++ program to:
    - Input 5 numbers into an array
    - Pass the array to a function
    - The function should calculate and return the **sum of elements**
    - Display the result in the main function

6. Write a C++ program to:
    - Input elements for a 2×2 matrix
    - Display the matrix in proper row and column format

7. Write a C++ program to:
    - Store elements in a 3×3 matrix
    - Print the matrix row by row

8. Write a C++ program to:
    - Input two 2×2 matrices
    - Add them
    - Display the resulting matrix

9. Write a C++ program to:
    - Input two 2×2 matrices
    - Multiply them
    - Display the result matrix

10. Write a C++ program to:

    Input elements for a 2×2 matrix

    Find the **transpose** of the matrix

    Display both the original matrix and the transposed matrix

    (Note: Transpose means converting rows into columns.)